WHMIS 2015

At Work



WorkSafeBC

About WorkSafeBC

At WorkSafeBC, we're dedicated to promoting safe and healthy workplaces across B.C. We partner with workers and employers to save lives and prevent injury, disease, and disability. When work-related injuries or diseases occur, we provide compensation and support injured workers in their recovery, rehabilitation, and safe return to work. We also provide no-fault insurance and work diligently to sustain our workers' compensation system for today and future generations. We're honoured to serve the workers and employers in our province.

Prevention Information Line

We provide information and assistance with health and safety issues in the workplace.

Call the information line 24 hours a day, 7 days a week to report unsafe working conditions, a serious incident, or a major chemical release. Your call can be made anonymously. We can provide assistance in almost any language.

If you have questions about workplace health and safety or the Occupational Health and Safety Regulation, call during our office hours (8:05 a.m. to 4:30 p.m.) to speak to a WorkSafeBC officer.

If you're in the Lower Mainland, call 604.276.3100. Elsewhere in Canada, call toll-free at 1.888.621.7233 (621.SAFE).

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Health and safety resources

You can find our health and safety resources on worksafebc.com, and many of them can be ordered from the WorkSafeBC Store at worksafebcstore.com.

In addition to books, you'll find other types of resources at the WorkSafeBC Store, including DVDs, posters, and brochures. If you have any questions about placing an order online, please contact a customer service representative at 604.232.9704 or toll-free at 1.866.319.9704.

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Introduction

Workers exposed to hazardous products may be at risk for many serious health problems, such as kidney or lung damage, sterility, cancer, allergic reactions, or burns. Some hazardous products can also cause fires or explosions.

The Workplace Hazardous Materials Information System (WHMIS) provides health and safety information about hazardous products in the workplace. Employers must use this information, as well as information specific to their workplaces, to educate and train workers to work safely with and near hazardous products.

This book explains the basics of WHMIS and answers some commonly asked questions. After reading this book, employers and workers should be able to understand how WHMIS information can help workers to work safely with and near hazardous products.

- Chapter 1 gives an overview of WHMIS and its goals, as well as the key changes from the original WHMIS to the updated WHMIS 2015. It briefly describes the four main elements of WHMIS 2015 (classification, labels, safety data sheets, and education and training) and the responsibilities of suppliers, employers, and workers.
- Chapter 2 explains how hazardous products are classified in WHMIS 2015, including descriptions of the new hazard groups, classes, and categories.
- Chapter 3 introduces the WHMIS 2015 pictograms and explains how they are used.
- Chapter 4 explains the requirements for WHMIS 2015 supplier and workplace labels.
- Chapter 5 describes the WHMIS 2015 safety data sheets (SDSs), including the information required on an SDS and the responsibilities of suppliers and employers.
- Chapter 6 describes WHMIS education and training requirements and the implementation of education and training programs.
- Chapter 7 includes tables outlining information on supplier labels and SDSs, and provides checklists for implementing WHMIS and education and training programs.
- Chapter 8 includes a list of resources for more WHMIS information.

1 What is WHMIS?

Overview of WHMIS

The Workplace Hazardous Materials Information System (WHMIS) provides Canadian workers with information about hazardous products used in the workplace. Under WHMIS, workers have the right to receive information about each hazardous product they use, handle, or store (for example, its identity, hazards, and safety precautions). The goal of WHMIS is to reduce injury and disease by communicating specific health and safety information about hazardous products to workers. Workers can use this information to reduce their exposure to hazardous products.

WHMIS gets an update

WHMIS first came into effect in 1988 through a series of complementary federal, provincial, and territorial laws and regulations. It was updated in early 2015 to reflect elements of a new initiative called the Globally Harmonized System of Classification and Labelling of Chemicals (GHS). Developed by the United Nations, GHS:

- Defines and classifies the hazards of chemical products
- Provides health and safety information on labels and safety data sheets, or SDSs (previously called material safety data sheets or MSDSs in the original WHMIS)

The goal of GHS is that the same set of rules for classifying hazardous products, and the same format and content for labels and SDSs, will be adopted and used around the world.

Currently, many countries have different systems for classifying and labelling hazardous products. Several different systems can exist even within the same country. This situation has been confusing for workers who need to understand the hazards of products in order to work safely. It's also been costly for companies that have to comply with many different systems. And it's been expensive for governments to regulate and enforce.

GHS has not replaced WHMIS. Instead, WHMIS has been aligned with GHS. This has led to many benefits, such as:

- Providing improved, consistent hazard information
- Encouraging the safe handling and use of hazardous products
- Promoting better emergency response
- Making it easier and less expensive for companies to comply
- Making trade easier
- Reducing the costs of regulation and enforcement

For the sake of clarity, the original WHMIS is now referred to as WHMIS 1988. The updated version is called WHMIS 2015.

Key changes from WHMIS 1988 to WHMIS 2015 include the following:

- Hazard classification criteria are more comprehensive. This improves the ability to indicate the severity of hazards.
- New hazard classes are included (for example, "Aspiration hazard").
- Physical hazard criteria are consistent with the Transportation of Dangerous Goods (TDG) Regulations.
- The language has been standardized.
- Supplier labels have a few new requirements (for example, the use of prescribed signal words, hazard statements, pictograms, and precautionary statements).
- SDSs follow a standard 16-section format with specific information requirements.

By late 2018, all suppliers and employers were required to comply with WHMIS 2015. WHMIS 1988 no longer applies.

Key elements of WHMIS 2015

GHS also defines an environmental hazards group. This group (and its classes) was not adopted in WHMIS 2015 since it is beyond the scope of WHMIS legislation (i.e., workplaces).

- **Classification**
  WHMIS 2015 divides hazardous products into two major hazard groups: physical hazards and health hazards.

  The physical and health hazard groups are split up into a number of classes. Some of the classes are divided even further into categories. The classes are depicted by pictograms (symbols surrounded by red, diamond-shaped borders) that identify their specific hazards. (For more information on classification, see page 14.) After a hazardous product has been classified, the following three WHMIS elements are used to communicate health and safety information.

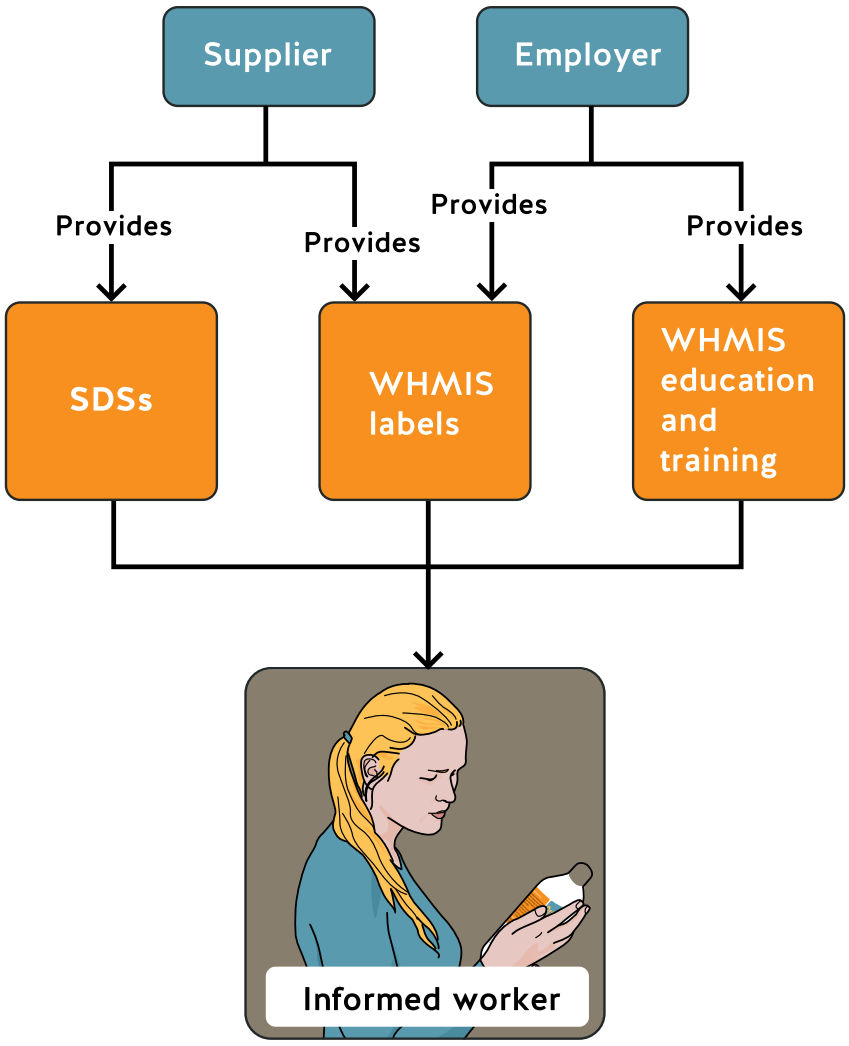
- **WHMIS labels**
  Labels on hazardous products alert workers to the identities of the products, their hazards, and precautions to be taken. Under WHMIS 2015, these labels must display some elements in a certain order. The hazard statements and precautionary statements have been standardized.

- **Safety data sheets (SDSs)**
  These documents provide detailed hazard and precautionary information. Under WHMIS 2015, SDSs use a 16-section format. The information required in each section has been standardized.

- **WHMIS education and training programs**
  Employers provide education and training for workers so that they can work safely with and near hazardous products. Workers need to know how WHMIS works, the hazards of hazardous products in their workplaces, and the safe work procedures they must follow.

Note: Not all hazardous products in the workplace are sold with WHMIS labels and safety data sheets. Some hazardous products are sold with labelling and hazard information meeting the requirements of other legislation. These products are either exempt or excluded from WHMIS requirements (see page 8).

How does WHMIS 2015 work?



Products covered by WHMIS 2015

The federal Hazardous Products Regulations (HPR) came into effect in early 2015 and replaced the Controlled Products Regulations. The HPR sets out the criteria for classifying hazards. If a product covered by the *Hazardous Products Act* meets the criteria to be included in a hazard class or category, it is considered to be a hazardous product. All hazardous products used in the workplace are covered by the WHMIS 2015 regulations, and a WHMIS program — including education and training — must be in place.

Products not covered by WHMIS 2015

The exclusions under WHMIS 2015 are:

- Explosives (as defined in the *Explosives Act*)
- Cosmetics, devices, drugs, or foods (as defined in the *Food and Drugs Act*)
- Pest control products (as defined in the *Pest Control Products Act*)
- Consumer products (as defined in the *Canada Consumer Product Safety Act*)
- Wood or products made of wood
- Nuclear substances, within the meaning of the *Nuclear Safety and Control Act*, that are radioactive
- Hazardous waste (being a hazardous product that is sold for recycling or recovery and is intended for disposal)
- Tobacco and tobacco products (as defined in the *Tobacco Act*)
- Manufactured articles

Many of these products are covered under other laws, as noted above. While a product may be exempt from the requirement to have a WHMIS label and SDS, employers must still provide education and training on health effects, safe use, and storage.

The rest of this book deals with hazardous products that require WHMIS supplier labels and SDSs (i.e., products that are not exempt from WHMIS 2015).

When consumer products are used in the workplace

Consumer products are those products that can be purchased in a store and are generally intended to be used in the home. They often include cleaning products, adhesives, or lubricants. These products are labelled according to other legislation. A comprehensive chemical safety program would include both hazardous products as regulated by WHMIS, and any other products that a worker may be exposed to (which includes consumer products). Workers should still receive education and training for safe use of these products.

Enforcement and inspections

WorkSafeBC enforces WHMIS in most workplaces in B.C. The federal Labour Program enforces WHMIS in federally regulated workplaces.

Inspectors have the authority to ensure that occupational health and safety legislation is being followed.

For WHMIS, for example, employers should be prepared to:

- Demonstrate that a WHMIS program is in place
- Show where the SDSs are for the hazardous products used at that workplace
- Show that hazardous products in use have the appropriate labels
- Show education and training records for employees who work with or may be exposed to a hazardous product

Inspectors may need to speak to workers to confirm that education and training have taken place. Workers should be able to answer these questions for every hazardous product they work with:

- What are the hazards of the product?
- How do I protect myself from those hazards?
- What do I do in case of an emergency?
- Where can I get further information?

Overview of responsibilities

In brief

The roles and responsibilities for suppliers, employers, and workers remain unchanged in WHMIS 2015.

The purpose of WHMIS is to reduce the likelihood of disease or injury in the workplace. WHMIS 2015 is the result of a collaborative effort between the federal, provincial, and territorial occupational health and safety agencies. It includes input from supplier, employer, and worker representatives.

WHMIS legislation exists at both the federal and provincial levels. **Federal legislation** (i.e., the *Hazardous Products Act* and the Hazardous Products Regulations) establishes which products are regulated under WHMIS 2015 and deals with either the importation or sale of these materials. Federal legislation defines suppliers as organizations or individuals who do any of the following in the course of business:

- Manufacture, sell, or distribute hazardous products in Canada
- Import hazardous products into Canada

Provincial legislation (in B.C., the *Workers Compensation Act* and the Occupational Health and Safety Regulation) covers the use of hazardous products in the workplace and identifies employers' responsibilities. Workers who work with or near hazardous products must know how to use, handle, and store them safely.

The following table summarizes the responsibilities of the various groups identified by WHMIS legislation. The roles and responsibilities of suppliers, employers, and workers have not changed in WHMIS 2015.

Table 2. Group responsibilities

Suppliers
• Ensure the appropriate classification of hazardous products. • Obtain or prepare up-to-date labels and SDSs for all hazardous products they sell or produce. • Provide these labels and SDSs to purchasers of hazardous products intended for use in a workplace.
Employers
• Educate and train workers on the hazards and safe use of hazardous products in the workplace. • Ensure that hazardous products are properly labelled. • Prepare workplace labels and SDSs (as necessary). • Ensure that up-to-date SDSs are readily available to workers. • Ensure effective control measures are in place to protect the health and safety of workers.
Workers
• Participate in WHMIS and chemical safety training programs. • Take necessary steps to protect themselves and their co-workers. • Participate in identifying and controlling hazards.
WorkSafeBC staff
• Administer WHMIS legislation. • Provide general information about WHMIS to employers and workers. • Ensure compliance with both federal and provincial WHMIS legislation.

Note that supplier labels and workplace labels are explained starting on page 33 and SDSs on page 48.

When an employer becomes a supplier

If employers import or produce a hazardous product, even if it is for their own use, they are considered to be the supplier of the hazardous product. This means they must provide an up-to-date SDS and attach a supplier label.

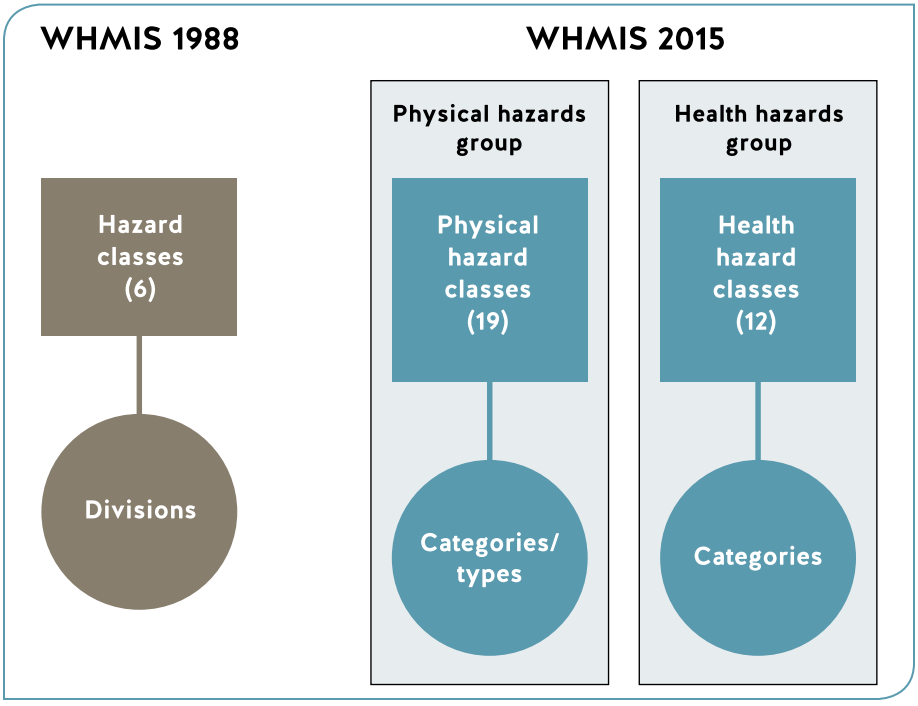
2 Classification of hazardous products

How hazardous products are classified in WHMIS 2015

The way that hazardous products are classified has changed in WHMIS 2015. Hazardous products are now divided into two **hazard groups**:

- **Physical hazards**, based on the physical or chemical properties of the product, such as flammability, reactivity, or corrosivity to metals
- **Health hazards**, based on the ability of the product to cause a health effect, such as:
  - Eye irritation
  - Respiratory sensitization (may cause allergy or asthma symptoms, or breathing difficulties)
  - Carcinogenicity (may cause cancer)

The two hazard groups are further divided into **hazard classes**. A brief listing of the hazard classes in each hazard group is shown on the next page, followed by an overview of the classes. Hazard classes are a way of grouping together products that have similar properties.



The figure above gives a high-level overview of how the WHMIS classification system has changed. In WHMIS 1988, controlled products were divided into hazard classes, which were further split into divisions. In WHMIS 2015, hazardous products are divided into two hazard groups. Each hazard group is divided into hazard classes. Each hazard class contains categories or types.

Physical hazards

The physical hazards group includes the following hazard classes:

- Combustible dusts
- Corrosive to metals
- Flammable aerosols
- Flammable gases
- Flammable liquids
- Flammable solids
- Gases under pressure
- Organic peroxides
- Oxidizing gases
- Oxidizing liquids
- Oxidizing solids
- Pyrophoric gases
- Pyrophoric liquids
- Pyrophoric solids
- Self-heating substances and mixtures
- Self-reactive substances and mixtures
- Simple asphyxiants
- Substances and mixtures which, in contact with water, emit flammable gases
- Physical hazards not otherwise classified

Note: Explosives are classified as physical hazards under GHS. And the "exploding bomb" pictogram appears in WHMIS 2015 because some hazardous products can explode. However, explosives are not included in WHMIS 2015 because they are covered by other legislation.

Health hazards

The health hazards group includes the following hazard classes:

- Acute toxicity
- Aspiration hazard
- Biohazardous infectious materials
- Carcinogenicity
- Germ cell mutagenicity
- Reproductive toxicity
- Respiratory or skin sensitization
- Serious eye damage/eye irritation
- Skin corrosion/irritation
- Specific target organ toxicity — single exposure
- Specific target organ toxicity — repeated exposure
- Health hazards not otherwise classified

Note: An environmental hazards group exists in GHS. This group (and its classes) was not adopted in WHMIS 2015 since it is beyond the scope of WHMIS legislation (i.e., workplaces). However, you may see the environmental classes listed on labels and SDSs. Including information about environmental hazards is allowed by WHMIS 2015.

Hazard classes

A hazardous product is a product that falls into one or more of the hazard classes described below. Suppliers classify these products and assign one or more of the appropriate pictograms (symbols or graphic images surrounded by borders; see page 23).

The following are only brief descriptions of each of the hazard classes.

Table 3. Overview of physical hazard classes

Physical hazard class	General description
Flammable gases; Flammable aerosols; Flammable liquids; Flammable solids	These four classes cover products that have the ability to ignite (catch fire) easily. The main hazards are fire or explosion.
Oxidizing gases; Oxidizing liquids; Oxidizing solids	These three classes cover oxidizers, which may cause or intensify a fire, or cause a fire or explosion.
Gases under pressure	This class includes compressed gases, liquefied gases, dissolved gases, and refrigerated liquefied gases. Compressed gases, liquefied gases, and dissolved gases are hazardous because of the high pressure inside the cylinder or container. The cylinder or container may explode if heated. Refrigerated liquefied gases are very cold and can cause severe cold (cryogenic) burns or injury.
Self-reactive substances and mixtures	These products may react on their own to cause a fire or explosion, or may cause a fire or explosion if heated.
Pyrophoric liquids; Pyrophoric solids; Pyrophoric gases	These products can catch fire very quickly (spontaneously) if exposed to air.
Self-heating substances and mixtures	These products may catch fire if exposed to air. These products differ from pyrophoric liquids or solids in that they will ignite only after a longer period of time or when in large amounts.
Substances and mixtures which, in contact with water, emit flammable gases	As the class name suggests, these products react with water to release flammable gases. In some cases, the flammable gases may ignite very quickly (spontaneously).

Physical hazard class	General description
Organic peroxides	These products are unstable, highly reactive, or explosive. They may cause a fire or explosion if heated.
Corrosive to metals	These products may be corrosive (chemically damaging or destructive) to metals.
Combustible dusts	This class is used to warn of products that are finely divided solid particles. If dispersed in air, the particles may catch fire or explode if ignited.
Simple asphyxiants	These products are gases that may displace (take the place of) oxygen in air and cause rapid suffocation.
Physical hazards not otherwise classified	This class is meant to cover any physical hazards that are not covered in any other physical hazard class. These hazards involve chemical reactions that result in serious injuries or deaths when the reactions occur. If a product is classified in this class, the hazard statement on the label and SDS will describe the nature of the hazard.

Table 4. Overview of health hazard classes

Health hazard class	General description
Acute toxicity	These products are fatal, toxic, or harmful if inhaled, if in contact with skin, or if swallowed. Acute toxicity refers to effects occurring following: • Skin contact or ingestion exposure to: – A single dose, or – Multiple doses given within 24 hours • An inhalation exposure of four hours Acute toxicity could result from exposure to the product itself. It could also result from a product that, upon contact with water, releases a gaseous substance that can cause acute toxicity.
Skin corrosion /irritation	This class covers products that cause severe skin burns (i.e., corrosion) and products that cause skin irritation.
Serious eye damage /eye irritation	This class covers products that cause serious eye damage (i.e., corrosion) and products that cause eye irritation.